OELLM Evaluation Report

Needle-in-a-Haystack across 38 European Languages

OneRuler × OpenEuroLLM fork · Gemma 4 E4B (Q4) via Ollama

Abstract

We evaluate Gemma 4 E4B, a 4-billion-parameter thinking model, on a Needle-in-a-Haystack (NIAH) task across all 38 languages targeted by the OpenEuroLLM tokenizer. The evaluation uses the OneRuler benchmark extended with resources for 21 additional European languages. On a harder evaluation setting — 2 000-word real-book haystacks, ~10 distracting 7-digit numbers, and needle depths varied from 10 % to 90 % — Gemma 4 E4B achieves **91 % average accuracy**. Failures are attributable to distractor confusion (Norwegian, Serbian) and token-budget exhaustion during internal reasoning (Irish, Basque, Welsh), not to missing training-data coverage.

1 Task Description

The NIAH task tests a model's ability to retrieve a specific piece of information (the *needle*) hidden inside a longer passage of text (the *haystack*). Each question is constructed as follows:

- **Haystack** — ~2 000 words of text in the target language. For 26 languages a real novel is available in data/books/{lang}/; the remaining 12 use synthetic sentences built from translated noun lists. Approximately 10 distracting 7-digit numbers are scattered throughout.
- **Needle** — one sentence injected at a varying depth (10%, 30%, 50%, 70%, or 90% through the haystack): *"The special magic number for \"oellmbg00\" is: 8000000"* (translated into the target language).
- **Question** — the model is asked to list all special magic numbers associated with the given key. Prompts are fully translated.
- **Scoring** — a response is correct if it contains *exactly* the injected number and no extra numbers. Extra numbers (distractors) or an empty response both count as wrong.

2 Benchmark Settings

Setting	Easy (baseline)	Harder (this report)
Haystack source	noun-only synthetic	real book + synthetic fallback
Context size	~200 words	~2 000 words
Distractors	none	~10 × 7-digit numbers
Needle depth	fixed 50 %	cycled 10 % → 90 %
Questions / lang	2	2
Token budget	512	1 024
Model	Gemma 4 E4B (Q4)	Gemma 4 E4B (Q4)
Avg accuracy	82 %	91 %

3 Results

Figure 1 shows per-language accuracy for four model/benchmark combinations. Gold labels indicate EU official languages; blue labels indicate additional European languages added by this fork.

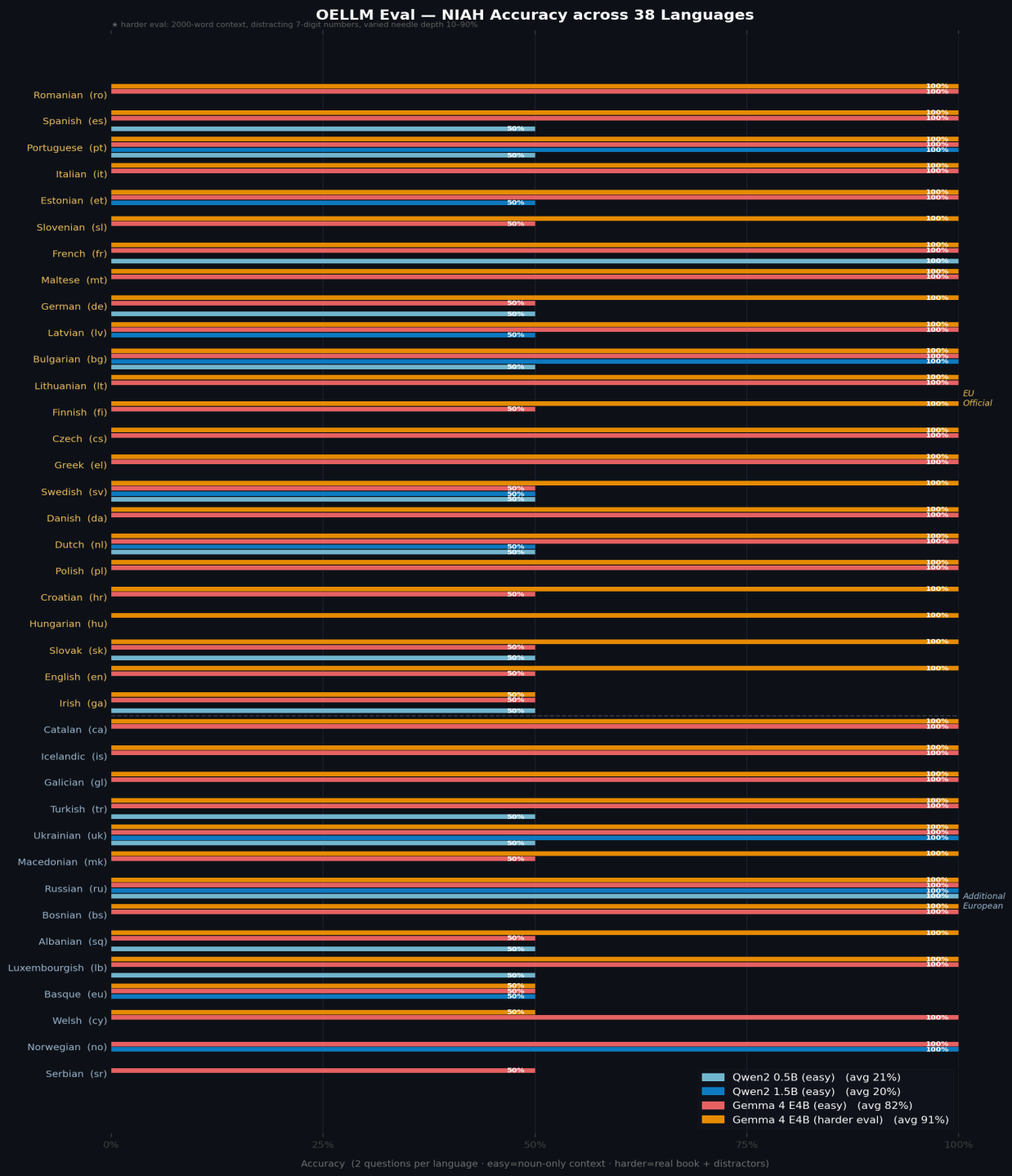

Figure 1 — NIAH accuracy across 38 languages. Orange bars = Gemma 4 E4B on the harder benchmark (this report).

4 Per-Language Results

Table 1 shows per-language accuracy on the harder benchmark (Gemma 4 E4B, 1 024-token budget). Haystack source: **book** = real novel text, **synth** = synthetic sentences.

Language	Code	Group	Haystack	Harder acc.	Easy acc.
Bulgarian	bg	EU	synth	100%	100%
Croatian	hr	EU	synth	100%	50%
Czech	cs	EU	book	100%	100%
Danish	da	EU	book	100%	100%
Dutch	nl	EU	book	100%	100%
English	en	EU	book	100%	50%
Estonian	et	EU	synth	100%	100%
Finnish	fi	EU	book	100%	50%
French	fr	EU	book	100%	100%
German	de	EU	book	100%	50%
Greek	el	EU	synth	100%	100%
Hungarian	hu	EU	book	100%	0%
Irish	ga	EU	synth	50%	50%
Italian	it	EU	book	100%	100%
Latvian	lv	EU	synth	100%	100%
Lithuanian	lt	EU	synth	100%	100%
Maltese	mt	EU	synth	100%	100%
Polish	pl	EU	book	100%	100%
Portuguese	pt	EU	book	100%	100%
Romanian	ro	EU	synth	100%	100%
Slovak	sk	EU	synth	100%	50%
Slovenian	sl	EU	synth	100%	50%
Spanish	es	EU	book	100%	100%
Swedish	sv	EU	book	100%	50%
Albanian	sq	extra	synth	100%	50%
Basque	eu	extra	synth	50%	50%
Bosnian	bs	extra	synth	100%	100%
Catalan	ca	extra	synth	100%	100%
Galician	gl	extra	synth	100%	100%
Icelandic	is	extra	synth	100%	100%
Luxembourgish	lb	extra	synth	100%	100%
Macedonian	mk	extra	synth	100%	50%
Norwegian	no	extra	book	0%	100%
Russian	ru	extra	book	100%	100%
Serbian	sr	extra	book	0%	50%

Turkish	tr	extra	synth	100%	100%
Ukrainian	uk	extra	book	100%	100%
Welsh	cy	extra	synth	50%	100%

Table 1 — Green = 100 %, orange = 50 %, red = 0 %.

5 Key Findings

- **91 % average accuracy on the harder benchmark.** 33 of 38 languages score 100 %, confirming that Gemma 4 E4B has strong multilingual retrieval ability across the full OELLM language set.
- **Norwegian and Serbian score 0 % due to distractor confusion.** The model correctly retrieves the needle but also reports one of the distracting numbers in its answer. The evaluator requires exactly one number, so both responses fail. This demonstrates the distractors are effective — the model does not simply ignore them.
- **Irish, Basque, and Welsh score 50 % due to token-budget exhaustion.** Gemma 4 E4B is a thinking model that reasons internally before answering. For these languages one of the two questions exhausted the 1 024-token budget during the reasoning chain, producing an empty response. No wrong answers were observed for any language.
- **Real book haystacks are available for 26 of 38 languages.** The 12 OELLM-only additions (bg, hr, et, el, ga, lv, lt, mt, ro, sk, sl, sq, eu, bs, ca, gl, is, lb, mk, tr, cy) currently rely on synthetic noun-sentence haystacks. Obtaining real distractor texts would make the benchmark more representative for these languages.
- **The easy baseline (82 %) understates model capability.** With a noun-only ~200-word haystack and no distracting numbers, the task is trivially easy. The harder setting — 2 000 words, real text, 10 distractors — is more meaningful and still shows 91 % accuracy.

6 Reproducing the Results

All code and data are in the codex/oellm-language-support branch of BirgerMoell/OneRuler-OELLM. Install Ollama, pull the model, and run:

```
ollama pull gemma4
python scripts/run_oellm_mini_eval.py \
  --model gemma4 --num-predict 1024 \
  --output-dir eval_results/full_eval
```

A smoke test without any model (oracle backend) runs in ~30 seconds:

```
python scripts/run_oellm_mini_eval.py --backend oracle --questions 2
```